

HEALTH AND MEDICINE

A subunit vaccine for multiple respiratory viruses

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Seasonal influenza viruses, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), and respiratory syncytial virus (RSV) drive substantial mortality worldwide. Vaccines against these respiratory viruses are traditionally administered separately, and, while some emerging technologies have shown promise to address multiple pathogens simultaneously, subunit protein vaccines have lagged behind. Here, we describe a nanoliposome-based vaccine platform that codisplays recombinant hemagglutinin ectodomains from three seasonal influenza strains, the SARS-CoV-2 receptor binding domain, and the RSV F ectodomain, in prefusion form. In mouse, ferret, and cotton rat models, immunization elicited protective immunity against the three major respiratory viruses, with antibody responses comparable to those induced by the corresponding monovalent formulations. These findings highlight the potential for multivalent recombinant protein vaccines delivered via nanoliposome carriers to provide immunologic protection against a broad spectrum of respiratory viruses.

INTRODUCTION

Circulating respiratory viruses such as seasonal influenza viruses, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), and respiratory syncytial virus (RSV) are major causes of morbidity and mortality worldwide. Globally, seasonal influenza viruses cause an estimated 3 to 5 million severe cases each year (1). RSV was estimated to cause roughly 5.2 million cases of RSV-related respiratory illness annually in older adults and medically vulnerable adults in industrialized countries (2). The COVID-19 pandemic is estimated to have caused ~14.8 million excess deaths globally in 2020–2021 (3). The “tripledemic” of seasonal influenza, COVID-19, and RSV was associated with ~1 million combined hospitalizations in the United States for the 2023–2024 respiratory virus season (4). Thus, there is a public health imperative to develop improved vaccines and increase vaccine coverage (5).

As of November 2024, US adult vaccination rates were ~35% for influenza, 18% for COVID-19, 40% for RSV in adults aged ≥75 years, and 32% for adults aged 60 to 74 years at increased risk for severe RSV (6). These coverage rates point toward the potential benefit of a single multivalent vaccine formulation that induces broad protection against multiple respiratory pathogens by reducing injection burden. Multivalent antigen strategies have been developed in vaccinology. In 2022, an mRNA vaccine encoding hemagglutinin (HA) from 20 influenza A virus subtypes and influenza B virus lineages showed promising preclinical results (7). Vaccines targeting other pathogens also use multivalent antigen approaches; for example, pneumococcal conjugate vaccines target multiple serotypes via polysaccharide-protein conjugates (8), and the

diphtheria-tetanus-acellular-pertussis-hepatitis-B-inactivated-poliovirus-Haemophilus influenzae-type-b (DTPa-HBV-IPV/Hib) vaccine combines multiple antigen types (9). For influenza, the Flublok vaccine incorporates recombinant HA antigens targeting three to four influenza strains (10). Moderna’s mRNA-1230 is an investigational mRNA vaccine encoding antigens from influenza (HA), SARS-CoV-2 (spike), and RSV (prefusion F), and it was evaluated in a phase 1 trial in 2023.

Still, relatively few multipathogen respiratory vaccines have been developed, particularly subunit vaccines, likely because of concerns regarding antigenic interference, manufacturing complexity, schedule mismatch, and regulatory hurdles. However, with biotechnology improvements in viral detection capability, accurate clinical trial testing of multirespiratory vaccine efficacy should be more feasible now than ever before, possibly using prevention of illness as an endpoint. Even so, the efficacy and manufacturability of a potent multipathogen respiratory vaccine are unclear. To address this, we made use of a cobalt-porphyrin-phospholipid (CoPoP) vaccine adjuvant system, which converts soluble antigens into a liposome-displayed format with strong immunogenicity (11). A COVID-19 vaccine based on CoPoP and the SARS-CoV-2 receptor binding domain (RBD) showed favorable immunogenicity and safety in phase 3 human trial testing (12). In this study, we included recombinant influenza HA ectodomains from World Health Organization (WHO)–recommended seasonal influenza A (H1N1 and H3N2) and influenza B strains (13), the SARS-CoV-2 RBD, and the ectodomain of RSV prefusion F to develop a single multivalent vaccine formulation to address these respiratory viral threats. We demonstrate that these protein antigens can be loaded simultaneously onto CoPoP nanoliposomes and elicit neutralizing antibodies, protecting against each of the respiratory viruses.

RESULTS

Antigen presentation and characterization of CoPoP, PHAD, and QS-21 (CPQ) liposomes

Liposomes were formed containing CoPoP alongside the immunostimulatory adjuvants PHAD and QS-21 to generate “CPQ” (Fig. 1A). Five recombinant antigens (characterized in fig. S1) were selected

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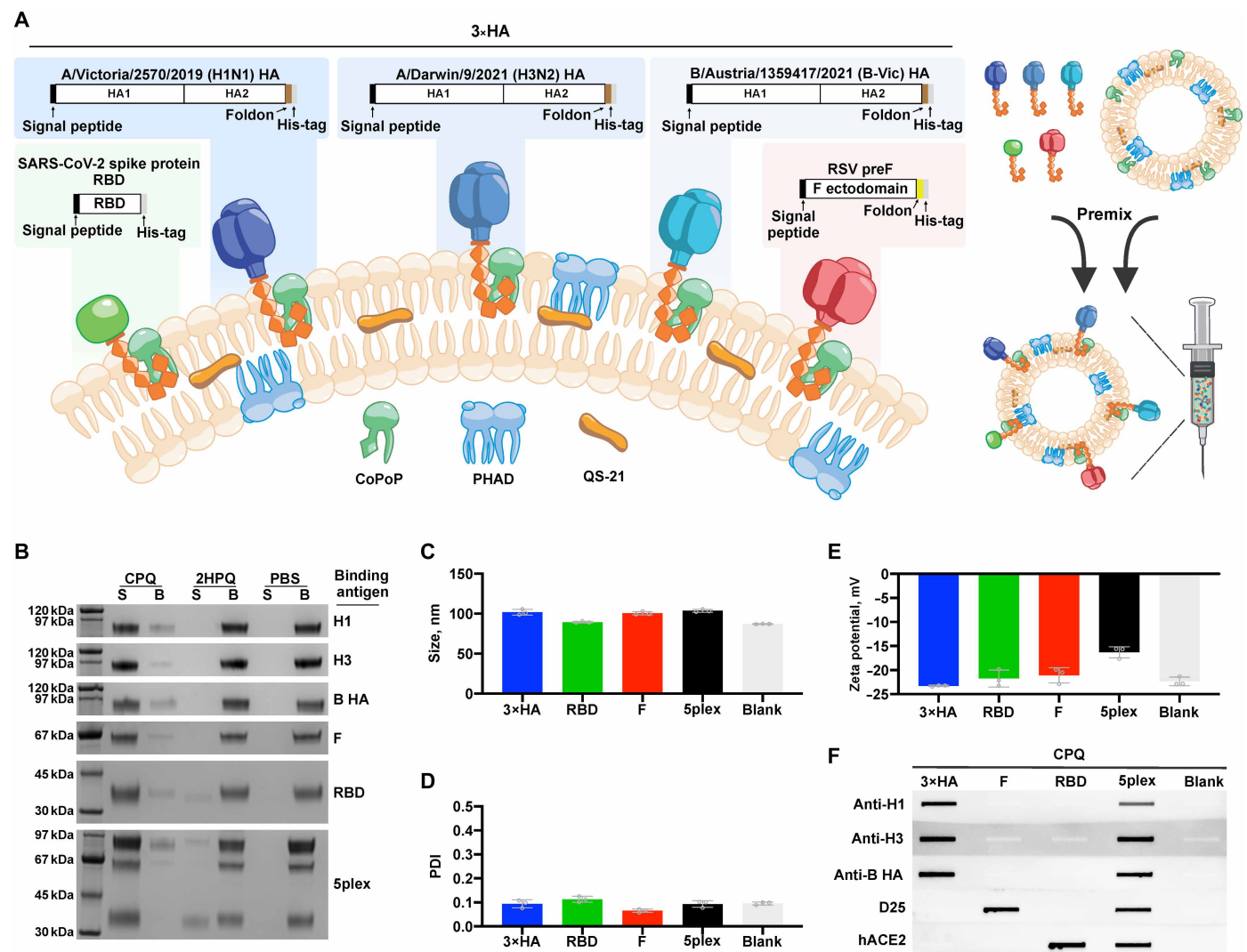


Fig. 1. Design and biophysical characterization of multiplex (5plex) CPQ liposomes. (A) Schematic of CPQ liposomes displaying His-tagged influenza HA ectodomain, His-tagged SARS-CoV-2 S protein RBD, and His-tagged recombinant RSV prefusion F ectodomain (DS2); antigens were premixed with the liposomes for 3 hours at room temperature to couple the antigens for vaccination. (B) Antigen conversion from soluble to liposome-bound format; “S” indicates supernatant, and “B” indicates Ni-nitrilotriacetic acid (NTA) bead fraction after magnetic separation. Biophysical characterization of (C) size, (D) polydispersity index (PDI), and (E) zeta potential of various CPQ formulations. (F) Slot blot assay showing detection of liposome-displayed antigens by monoclonal antibodies (mAbs). In [(C), (D), and (E)], data are presented as means $\pm$ SD.

with C-terminal His-tags, which enable chelation with cobalt and display on CoPoP liposome surfaces. Recombinant HA ectodomain of A/Victoria/2570/2019 (H1N1), A/Darwin/9/2021 (H3N2), and B/Austria/1359417/2021 (B/Victoria lineage; B-Vic) was produced on the basis of the recommended composition of influenza virus vaccines during the 2022–2023 Northern hemisphere influenza season by the WHO as described (13, 14). Recombinant SARS-CoV-2 RBD (residues 319 to 541, C-terminal His-tag) was derived from transfected human embryonic kidney (HEK) 293 cells (15). The RSV F ectodomain with prefusion stabilizing mutations, as a second-generation “DS2” stabilized immunogen, derived from the RSV-A Long strain, was used (16). After room temperature incubation of the antigens with CPQ, the mixture was incubated with Ni-nitrilotriacetic acid (NTA) magnetic beads in the competitive binding assay, followed by bead separation and SDS-polyacrylamide gel electrophoresis (PAGE)

analysis to assess CPQ binding (Fig. 1B). A control liposome, 2HPQ, in which CoPoP was replaced with cobalt-free porphyrin-phospholipid (PoP), was also included. The five antigens were able to efficiently bind to CPQ, given that most bands appeared in the magnetic bead supernatant (“S”) lane. A small fraction of RBD remained associated with 2HPQ liposomes, possibly due to electrostatic association. The liposome sizes were around 100 nm (Fig. 1C) with a polydispersity index less than 0.2 in phosphate-buffered saline (PBS) of individual CPQ and multiplex CPQ (5plex) (Fig. 1D) and all formulations showed a negative zeta potential (Fig. 1E). Slot blot results showed that corresponding monoclonal antibodies (mAbs) recognized each of the three HA, F, or RBD antigens, indicating the accessible presentation of antigens on the bilayer surface of CPQ (Fig. 1F). A fluorescent quenching assay was performed as an additional confirmation of the particle-forming kinetics of a fluorophore-labeled RBD antigen, which

showed ~80% quenching after 1 hour of incubation (fig. S2). Quantitative enzyme-linked immunosorbent assays (ELISAs) were able to confirm both the expected total antigen concentration and the amount of unbound antigen present in the 5plex CPQ formulation, using H1 HA as a representative example (fig. S3).

5plex CPQ immunization elicits neutralizing antibodies against influenza, SARS-CoV-2, and RSV

Outbred Institute of Cancer Research (ICR) mice were immunized with CPQ liposomes displaying either three HA ectodomains (3×HA), F, RBD, or all five antigens, followed by a booster with the same antigen dose 3 weeks after prime and sera collection on day 47 (Fig. 2A). ELISAs were performed to measure binding immunoglobulin G (IgG) antibody levels against H1 (Fig. 2B), H3 (Fig. 2D), or B-HA (Fig. 2F) ectodomain. ELISA titers were determined by serial dilution, with representative data shown in fig. S4. Both the 3×HA and 5plex CPQ formulations elicited strong IgG responses. Notably,

under the same prime-boost regimen and antigen doses, the single-antigen and 5plex formulations elicited similar antigen-specific antibody responses. The neutralizing antibody titers against influenza were also analyzed with the hemagglutination inhibition (HAI) assay, an essential metric for influenza vaccine functional immunogenicity. The 3×HA and 5plex CPQ immunizations induced potent HAI titers against A/California/04/2009 (H1N1) (Fig. 2C), A/Switzerland/9715293/2013 (H3N2) (Fig. 2E), and B/Washington/02/2019 (B/Victoria lineage) (Fig. 2G), confirming the elicitation of neutralizing antibodies.

The monovalent RBD CPQ formulation and 5plex CPQ immunization also induced potent binding IgG antibodies measured by ELISA (Fig. 2H). Neutralizing activity was assessed using a pseudovirus (PsV)-based neutralization assay with a luciferase reporter. Both RBD CPQ and 5plex CPQ immunization elicited high neutralizing antibody titers against SARS-CoV-2 PsV particles (Fig. 2I). To confirm RBD-specific antibody activity, sera were evaluated using the

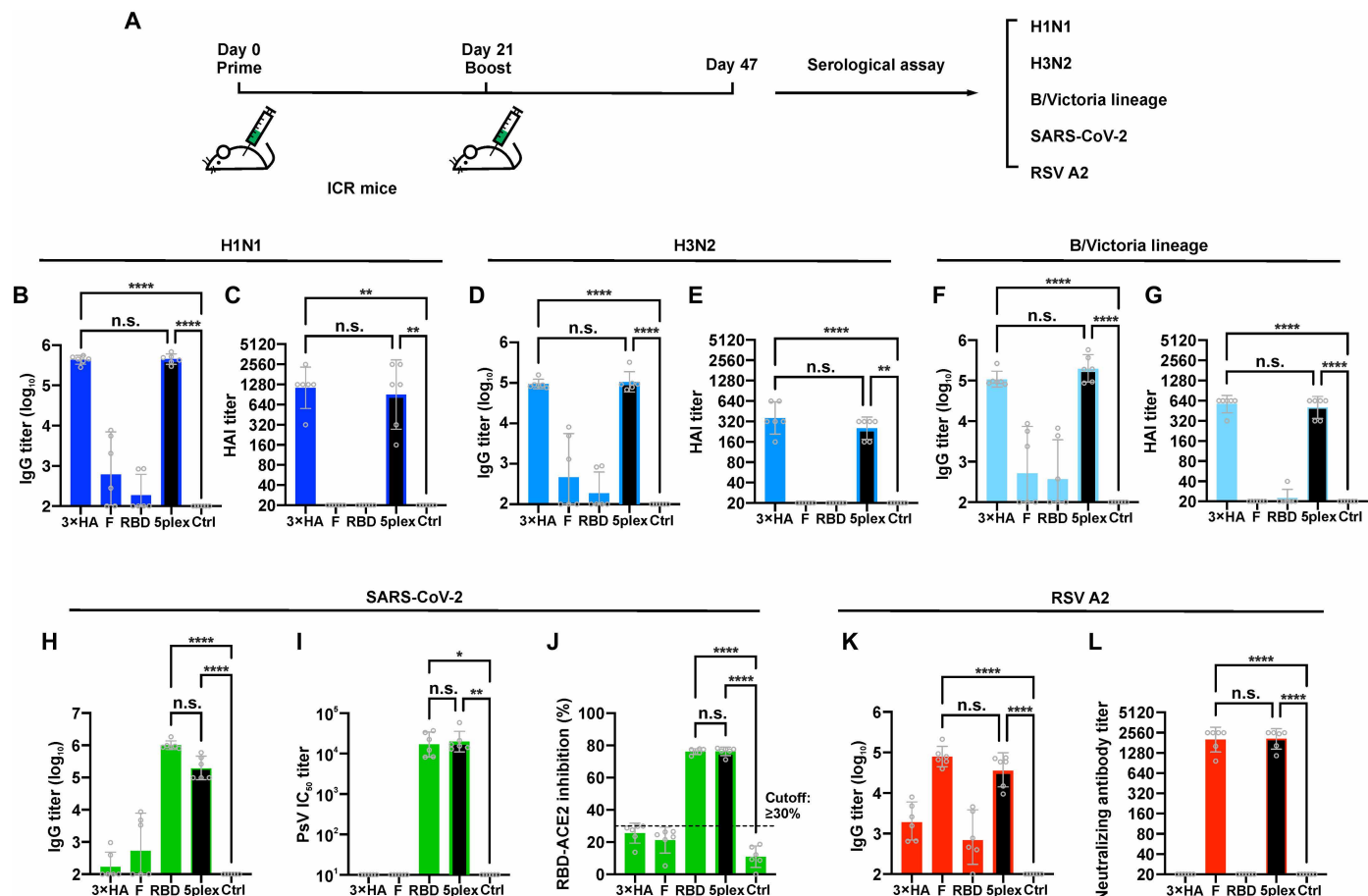


Fig. 2. Serological comparison of CPQ liposome formulations containing individual or 5plex antigens. (A) Outbred ICR mice received two immunizations 3 weeks apart, followed by serum collection on day 47 after initial immunization; $n = 6$. (B) IgG titers against recombinant HA ectodomain without the trimerization domain of A/Brisbane/02/2018 (H1N1). (C) HAI titers against A/California/04/2009 (H1N1). (D) IgG titers against recombinant HA ectodomain without the trimerization domain of A/Hong Kong/2671/2019 (H3N2). (E) HAI titers against A/Switzerland/9715293/2013 (H3N2). (F) IgG titers against recombinant HA ectodomain without the trimerization domain of B/Washington/02/2019 (B/Victoria lineage). (G) HAI titers against B/Washington/02/2019 (B/Victoria lineage). (H) IgG titers against SARS-CoV-2 RBD. (I) SARS-CoV-2 pseudovirus (PsV) 50% inhibitory concentration (IC_{50}) titers. (J) RBD-ACE2 inhibition; a cutoff $\geq 30\%$ was used to determine the presence of SARS-CoV-2 neutralizing antibodies in the sample. (K) IgG titers against RSV A2 F ectodomain. (L) Neutralizing antibody titers against RSV A2. In [(B) to (I), (K), and (L)], the data are presented as the geometric mean with 95% confidence interval. In (J), data are presented as means $\pm$ SD. Significance was analyzed by a one-way analysis of variance (ANOVA) followed by Tukey's multiple comparisons test. * $P < 0.05$; ** $P < 0.01$; **** $P < 0.0001$; n.s., not significant.

SARS-CoV-2 surrogate virus neutralization test (sVNT), which measures inhibition of horseradish peroxidase (HRP)–conjugated RBD binding to plate-coated angiotensin-converting enzyme 2 (ACE2). Antibodies induced by immunization blocked RBD–ACE2 interaction, consistent with the established mechanism of protection in which SARS-CoV-2 neutralizing antibodies prevent viral entry (Fig. 2J).

RSV fusion glycoprotein was used as the ELISA coating antigen, and the binding IgG titers demonstrated that both F CPQ and 5plex CPQ immunization induced a strong RSV F binding antibody response (Fig. 2K). Consistent with the ELISA results, neutralizing antibody titers measured against RSV A2 showed that both F CPQ and 5plex CPQ immunization induced neutralizing antibodies (Fig. 2L).

We evaluated the immunogenicity of multiple contemporary adjuvants in comparison with the 5plex CPQ formulation by formulating an identical antigen dose of 0.6 μ g of each antigen and administering the same immunization regimen described in Fig. 2A. HAI titers against H1N1, H3N2, and B/Victoria strains showed that CPQ resulted in higher or comparable functional antibody compared to the non-particle-forming 2HPQ control adjuvant, AddaVax, AS01, and Alum adjuvants (fig. S5). Notably, CPQ formulation induced higher functional antibody titers measured by PsV and sVNT test compared to all adjuvants, consistent with enhanced antigen presentation resulting from surface display of the RBD on CPQ liposomes. CPQ formulation demonstrated comparable immunogenicity compared to 2HPQ and AS01, whereas stronger immunogenicity compared to AddaVax and Alum, against RSV A2 and RSV B (fig. S5).

As no measurable reduction in antigen-specific antibody responses was observed in the 5plex formulation at the 0.6- μ g dose level (Fig. 2), immunogenicity was also evaluated at a lower antigen dose of 0.1 μ g for each antigen. As shown in fig. S6, there was no deleterious IgG titers difference between the 3 $\times$ HA, RBD, or F CPQ formulations compared to the 5plex CPQ formulation, indicating that the 5plex CPQ formulation did not result in a measurable reduction in antigen-specific antibody responses under the conditions tested.

Under reduced antigen dose conditions, we observed that, although total IgG titers were lower, cross-reactive nonspecific IgG titers were also reduced in the group that received foldon-containing antigens, particularly for the three recombinant HAs and recombinant RSV F. To investigate the source of low-level, nonspecific antibody titers in Fig. 2B, we performed an ELISA using an irrelevant His-tagged protein (Pfs48/45), which does not contain a foldon domain, as the coating antigen (17). Sera from immunized groups showed minor cross-reactivity to the irrelevant His-tagged protein, suggesting that some antibodies may have been directed against the His-tag. However, off-target antibody levels were $\sim$ 3 orders of magnitude lower compared to CPQ Pfs48/45 postvaccination sera, suggesting that most of the total IgG was directed against antigens and not their His-tags (fig. S7). In phase 2 human trial testing of a CoPoP RBD vaccine, participants did not elicit significant anti-His-tag antibodies (18).

3 $\times$ HA CPQ and 5plex CPQ conferred protection in mice and ferrets against influenza infection

BALB/c mice were immunized with 3 $\times$ HA CPQ or 5plex CPQ formulations containing 0.6 μ g of each antigen. Mice received the same formulation as the booster on day 21, followed by intranasal inoculation of mouse-adapted influenza virus on day 42 (Fig. 3A). Before influenza challenge, ELISA showed that both 3 $\times$ HA CPQ and 5plex CPQ immunization induced robust binding IgG titers against the HA ectodomains of three influenza strains (Fig. 3B). All mice challenged

with mouse-adapted A/California/04/2009 (H1N1) or mouse-adapted B/Malaysia/2506/2004 (B/Victoria lineage) survived to day 14, indicating that 3 $\times$ HA CPQ and 5plex CPQ conferred complete protection against these two influenza strains (Fig. 3C). In addition, no substantial weight loss (Fig. 3D) or increased clinical scores were observed (fig. S8) among mice that received 3 $\times$ HA CPQ or 5plex CPQ and were challenged with mouse-adapted A/California/04/2009 (H1N1) or mouse-adapted B/Malaysia/2506/2004 (B/Victoria lineage). However, one mouse from the 5plex CPQ immunization group succumbed on day 6 after challenge with the mouse-adapted A/Hong Kong/1/1968 (H3N2), while all mice survived after receiving 3 $\times$ HA CPQ vaccination, and overall decreases in body weight and increases in clinical scores were observed. The reduced efficacy during early infection was likely associated with antigenic differences between the recombinant HA ectodomain of A/Darwin/9/2021 (H3N2) immunogen and that of the mouse-adapted A/Hong Kong/1/1968 (H3N2) challenge strain (table S1).

To further evaluate the protective efficacy of the 5plex CPQ against H1N1 virus infection, we immunized three male and three female ferrets per group with two doses of 5plex CPQ formulation, including 6 μ g of each of the five antigens. Ferrets received two identical doses 3 weeks apart and were challenged with A/Wisconsin/67/2022 (H1N1) influenza virus on day 42 (Fig. 3E). The recombinant influenza vaccine Flublok was included as a comparator vaccine, administered at the same per-HA doses in ferrets and following the same experimental schedule. While 5plex CPQ immunization induced significantly higher IgG titers detected in sera collected on day 39 (Fig. 3F), similar to what we observed in ICR mice immunization (fig. S8), HAI titers showed that both 5plex CPQ and Flublok induced comparable neutralizing antibody levels on day 39 after prime, and significantly higher antibody levels were detected in 5plex CPQ-immunized groups on day 49, 7 days postinfection (dpi), compared to the Flublok-immunized group and the untreated group (Fig. 3G). In addition, both the 5plex CPQ- and Flublok-immunized groups showed significantly lower infectious titers [50% tissue culture infectious dose (TCID₅₀)/ml] than the untreated group at 5 days postinfection, indicating that 5plex CPQ conferred protection comparable to that of Flublok (Fig. 3H). 5plex CPQ immunization elicited neutralizing antibody responses against A/Wisconsin/67/2022 (H1N1) that were comparable to those induced by the licensed Flublok vaccine, suggesting that 5plex CPQ can match commercial benchmarks (Fig. 3I).

RBD CPQ or 5plex CPQ conferred protection in K18-hACE2 transgenic mice against SARS-CoV-2 infection

K18-hACE2 transgenic mice were immunized with 0.6 μ g of RBD alone or each of the five antigens in the 5plex formulation in a two-dose regimen before the rSARS-CoV-2–Nluc (WA1) challenge, on the basis of previously developed experimental protocols (Fig. 4A). We first performed ELISAs with sera collected on day 28 against each of the five antigens (Fig. 4B), and the results suggested that the immunization induced corresponding binding IgG antibodies. Infection with rSARS-CoV-2–Nluc (WA1) virus caused substantial body weight loss starting from 2 days postinfection and became fatal at 6 days postinfection in the control group, while RBD CPQ and 5plex CPQ groups were protected from weight loss (Fig. 4C). The plaque reduction microneutralization tests (PRMNTs) were also performed with sera collected 2 weeks after prime and 2 weeks after boost, before challenge with rSARS-CoV-2–Nluc (WA1) (fig. S9). The results indicated the presence of neutralizing antibodies in both RBD CPQ

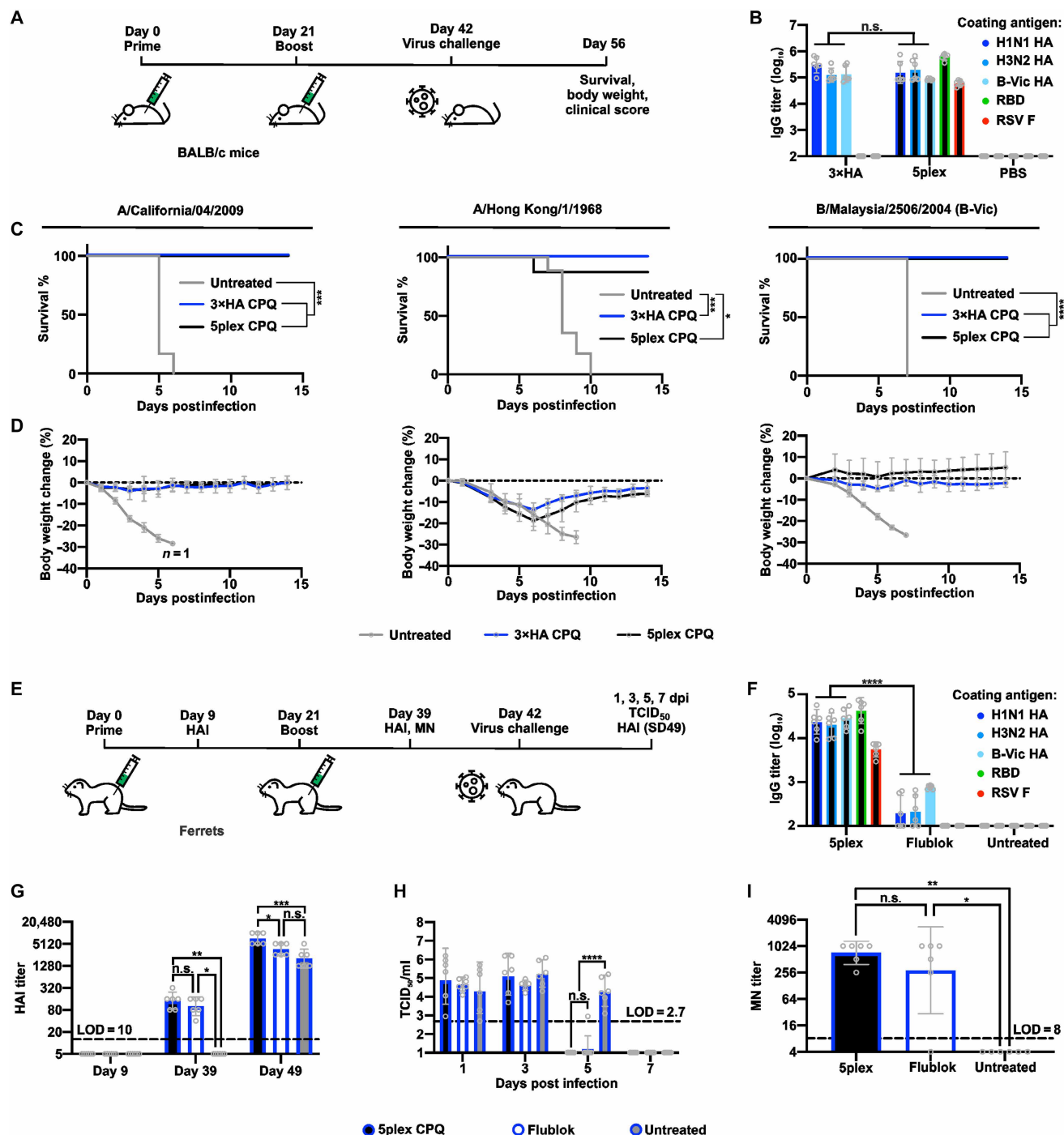


Fig. 3. 5plex CPQ immunization conferred protection against influenza infection. (A) BALB/c mice received two doses of 3×HA CPQ or 5plex CPQ containing 0.6 µg of each antigen 3 weeks apart and then challenged with 1.2×10^5 PFU ($2.5 \times \text{LD}_{50}$) of mouse-adapted A/California/04/2009 (H1N1) ($n = 6$), 10 PFU ($10 \times \text{LD}_{50}$) of mouse-adapted A/Hong Kong/1/1968 (H3N2) ($n = 6$), or 3.2×10^6 PFU ($10 \times \text{LD}_{50}$) of mouse-adapted influenza B/Malaysia/2506/2004 (B/Victoria lineage) ($n = 6$ for the treated group and $n = 5$ for the untreated group) on day 42. (B) ELISAs were performed with sera collected from BALB/c mice before A/California/04/2009 (H1N1) influenza virus challenge on day 42, and the IgG titers against each of the five antigens: recombinant HA ectodomain without trimerization domain of A/Brisbane/02/2018 (H1N1) (H1N1 HA), A/Hong Kong/2671/2019 (H3N2) (H3N2 HA), B/Washington/02/2019 (B/Victoria lineage) (B-Vic HA), SARS-CoV-2 RBD, and RSV A2 F ectodomain (RSV F); $n = 6$. (C) Survival. (D) Average body weight change. (E) Ferrets received two doses of 5plex CPQ containing 6 µg of each of the five antigens or Flublok administered at the same per-HA dose, 3 weeks apart, and were challenged with A/Wisconsin/67/2022 (H1N1); $n = 6$. (F) IgG titers measured with sera collected on day 39 after prime against the antigen described in (B). (G) HAI titers on days 9, 39, and 49 after prime. (H) TCID₅₀ infectious titers at 1, 3, 5, and 7 days postinfection (dpi). (I) Microneutralization (MN) titers tested with sera collected on day 39 after prime against A/Wisconsin/67/2022 (H1N1). In (D), data are presented as means $\pm$ SD. In [(B), (F), (G), (H), and (I)], data are presented as geometric mean with 95% confidence interval. Significance in [(B), (F), (G), and (H)] was analyzed by a one-way ANOVA followed by Tukey's multiple comparisons test. Significance in (C) was analyzed by the log-rank test. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$; n.s., not significant.

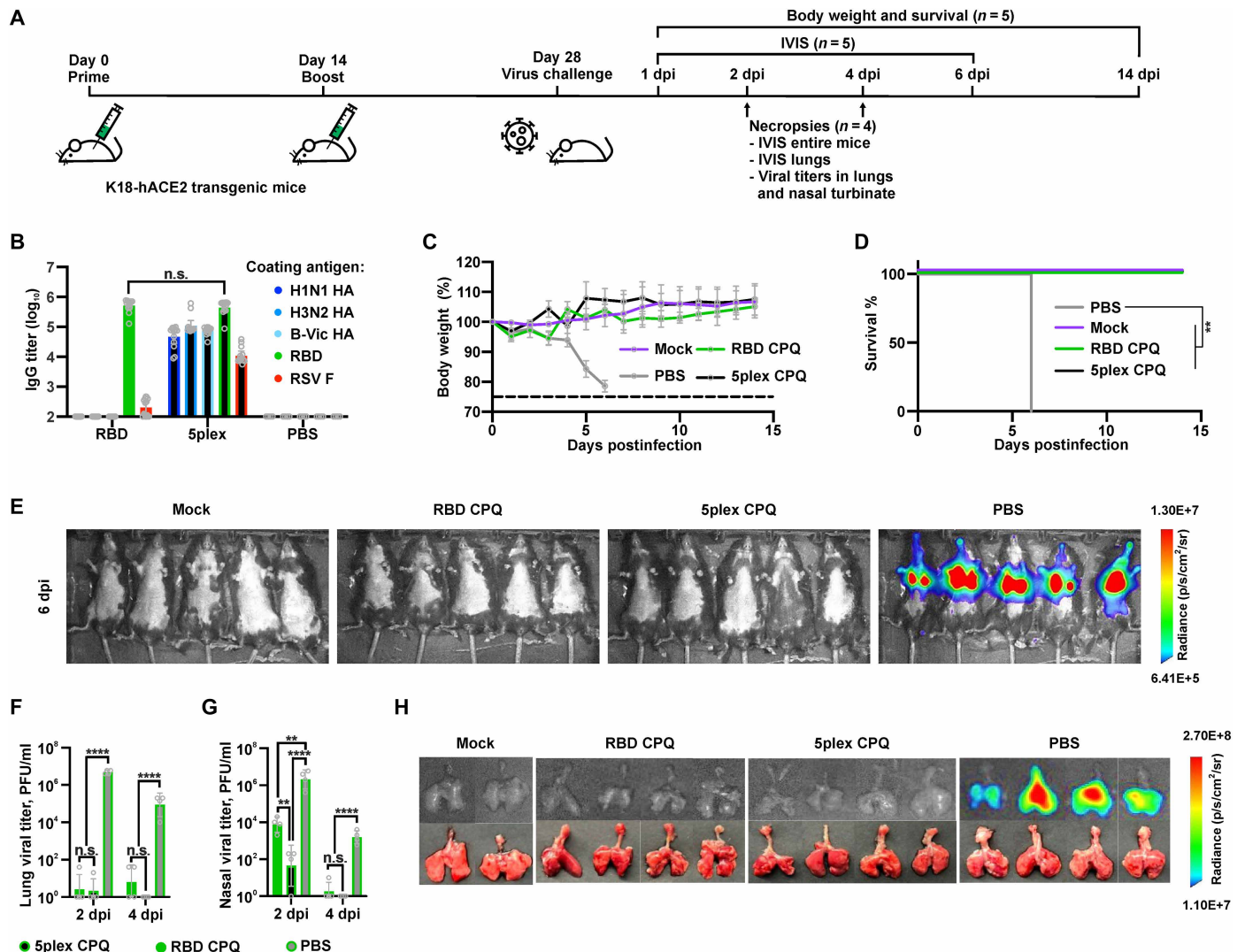


Fig. 4. 5plex CPQ immunization conferred protection against SARS-CoV-2 infection. (A) Immunization and experiment design. IVIS refers to in vivo imaging system. (B) Prechallenge IgG titers against recombinant HA ectodomain without trimerization domain of A/Brisbane/02/2018 (H1N1) (H1N1 HA), A/Hong Kong/2671/2019 (H3N2) (H3N2 HA), B/Washington/02/2019 (B/Victoria lineage) (B-Vic HA), SARS-CoV-2 RBD, and RSV A2 F ectodomain (RSV F); $n = 13$ for the RBD and 5plex groups and $n = 4$ for the PBS group. (C) Body weight change. (D) Survival of mice post-rSARS-CoV-2-Nluc (WA1) challenge; $n = 5$. (E) In vivo luciferase signal monitored at 6 days postinfection (dpi). (F) Virus recovered from lung homogenates at 2 and 4 days postinfection; $n = 4$. (G) Virus recovered from nasal turbinates at 2 and 4 days postinfection; $n = 4$. (H) Ex vivo Nluc signal detected in lungs at 2 days postinfection; $n = 4$, and $n = 2$ for the mock group. In (B), data are presented as the geometric mean with 95% confidence interval. In [(C), (F), and (G)], data are presented as means $\pm$ SD. Significance in [(B), (F), and (G)] was analyzed by a one-way ANOVA followed by Tukey's multiple comparisons test. Significance in (D) was analyzed by the log-rank test. ** $P < 0.01$; **** $P < 0.0001$; n.s., not significant.

and 5plex CPQ groups. All unvaccinated mice reached humane endpoint criteria by 6 days postinfection, whereas RBD CPQ and 5plex CPQ protected mice against lethal rSARS-CoV-2-Nluc (WA1) challenge (Fig. 4D). In addition, we monitored the in vivo viral distribution of rSARS-CoV-2-Nluc (WA1) based on the luciferase activity. No luciferase signal was detected at 1, 2, 4 (fig. S9), and 6 days postinfection (Fig. 4E), indicating reduced viral burden in the RBD CPQ and 5plex CPQ immunization groups, while infection progressed from 1 day postinfection in the PBS group.

Next, necropsy examinations were performed, and both lung viral titers and NanoLuc (Nluc) reporter signal intensities were quantified at 2 and 4 days postinfection. A strong Nluc signal was detected in the lungs of the PBS group at 2 days postinfection (Fig. 4H) and

4 days postinfection (fig. S9). Consistently, significantly lower viral titers (Fig. 4F) and Nluc signal (fig. S9) were detected in lung homogenates collected at 2 and 4 days postinfection, suggesting the robust protection conferred by RBD CPQ or 5plex CPQ immunization.

Nasal turbinate viral titers and Nluc reporter signal intensities were also quantified. Both Nluc signal and viral titers were detected in the RBD CPQ- and 5plex CPQ-vaccinated groups but were significantly lower than those in the PBS group, at 2 and 4 days postinfection (Fig. 4G and fig. S9). In addition, residual nasal viral titers were not associated with body weight loss or mortality in the RBD CPQ- and 5plex CPQ-vaccinated groups, and neither infectious viral titers nor Nluc signal was detected in lung homogenates. Residual nasal infection may reflect the intramuscular route of immunization,

which primarily induced robust circulating RBD-specific IgG rather than strong mucosal immunity in the upper respiratory tract.

F CPQ and 5plex CPQ conferred protection in cotton rats against RSV A2 infection

Last, we immunized 6- to 8-week-old cotton rats with 1.8 μ g of F or 1.8 μ g of each of the five antigens on days 0 and 21 (Fig. 5A). The alum-adsorbed formalin-inactivated RSV vaccine (FI-RSV) was also administered using the same schedule. FI-RSV was developed

in the 1960s as one of the earliest efforts of RSV prevention but resulted in severe enhanced respiratory disease (ERD) when vaccinated patients encountered natural RSV infection, leading to hospitalizations and fatalities, and left challenges for the RSV vaccine development (19).

ELISA titers against each of the five antigens indicated that F CPQ and 5plex CPQ induced antibodies against the corresponding antigen (Fig. 5B). Binding IgG was also detected by ELISA against the RSV A2 in the F CPQ, 5plex CPQ, and FI-RSV groups on days 0, 21,

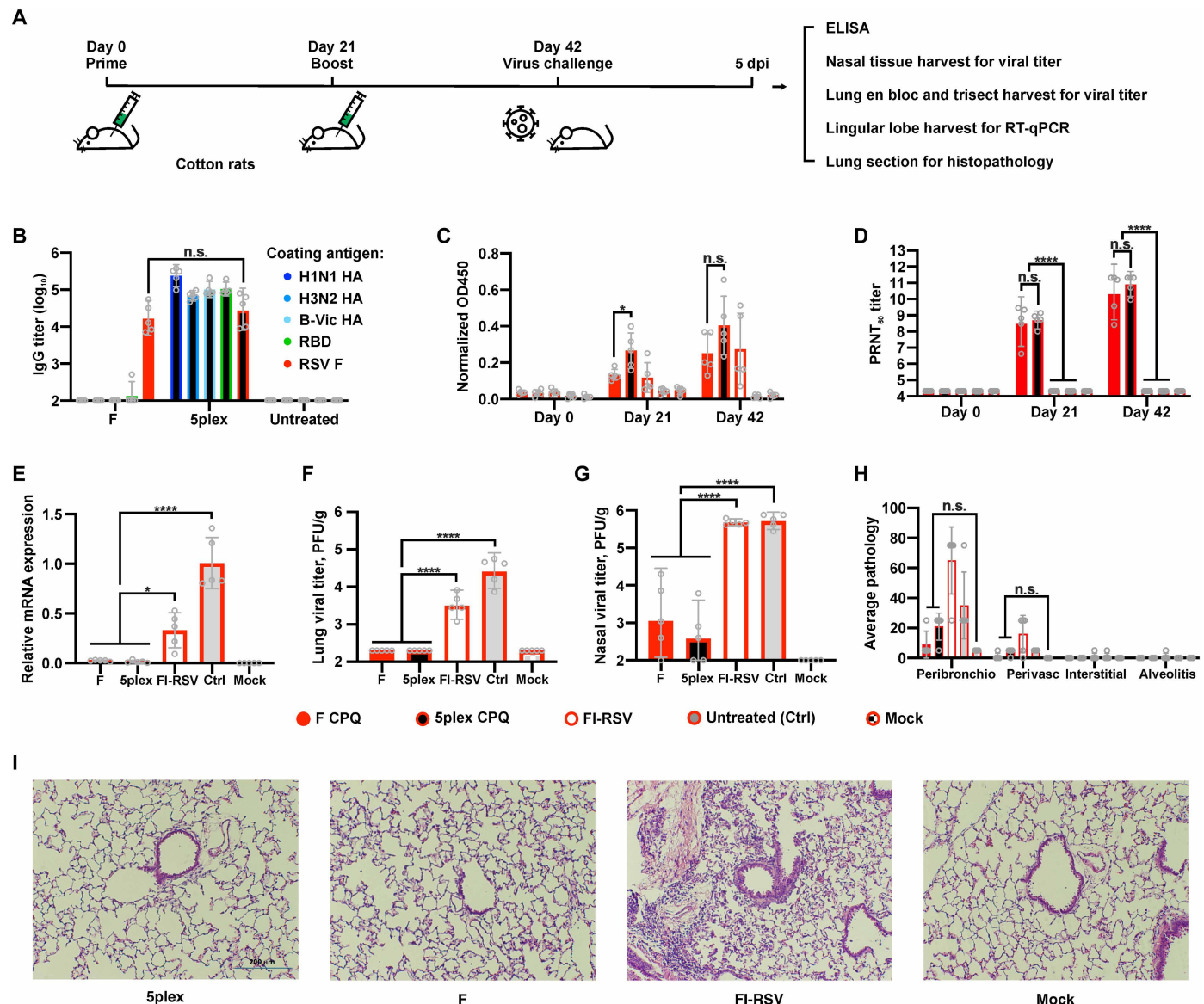


Fig. 5. 5plex CPQ immunization conferred protection against RSV A2 infection. (A) Immunization scheme and experiment design for cotton rat in vivo challenge; $n = 5$. RT-qPCR, reverse transcription quantitative polymerase chain reaction. (B) ELISAs were performed with sera collected on day 42 against recombinant HA ectodomain without trimerization domain of A/Brisbane/02/2018 (H1N1) (H1N1 HA), A/Hong Kong/2671/2019 (H3N2) (H3N2 HA), B/Washington/02/2019 (B/Victoria lineage) (B-Vic HA), SARS-CoV-2 RBD, and RSV A2 F ectodomain (RSV F). (C) Normalized OD₄₅₀ for IgG against purified F protein of sera collected on days 0, 21, and 42. (D) A plaque reduction neutralization test with a 60% reduction endpoint (PRNT₆₀) was used to measure RSV neutralizing antibodies in sera collected on days 0, 21, and 42. (E) RSV A2 viral mRNA expression level in lung homogenates harvested at 5 days postinfection; expression levels were normalized to β -actin. (F and G) RSV A2 viral titers detected in lung (F) and nasal homogenates (G) at 5 days postinfection. (H) Average lung pathology score at 5 days postinfection. (I) Pulmonary histopathology at 5 days postinfection; scale bar, 200 μ m. In [(B), (D), (F), and (G)], data are presented as the geometric mean with 95% confidence interval. In [(C), (E), and (H)], the data are presented as means $\pm$ SD. Significance was analyzed by a one-way ANOVA followed by Tukey's multiple comparisons test. * $P < 0.05$; **** $P < 0.0001$; n.s., not significant.

and 42 (Fig. 5C). However, only F CPQ or 5plex CPQ immunization was able to elicit neutralizing antibodies against RSV, as determined by the plaque reduction neutralization test, whereas FI-RSV immunization induced ineffective neutralizing activities (Fig. 5D). This is consistent with the known effects of formalin inactivation, which disrupts the prefusion F conformation and diminishes neutralization-sensitive epitopes, thereby reducing protective antibody responses. Reverse transcription quantitative polymerase chain reaction (PCR) also showed a significant reduction of viral mRNA in lung samples from the F CPQ- or 5plex CPQ-immunized groups while showing detectable viral mRNA remained after FI-RSV immunization (Fig. 5E).

Next, we analyzed the viral titers in lung and nasal homogenates. After two doses of F CPQ or 5plex CPQ immunization, no detectable viral titers were observed in lung homogenates (Fig. 5F). However, detectable viral titers remained in nasal homogenates (Fig. 5G). The difference between lung and nasal viral titers was similar to that observed in SARS-CoV-2 in vivo challenge, suggesting that intramuscular immunization primarily induced systemic immune response with strong circulating neutralizing antibodies, which effectively controlled viral replication in deeper organs such as the lung.

We analyzed the pulmonary histopathology in lung tissue collected from the F CPQ-, 5plex CPQ-, and FI-RSV-immunized groups and compared them to the unvaccinated control group. Four pathological indicators were analyzed: peribronchiolar inflammation ("peribronchio"), perivascular inflammation ("perivasc"), interstitial inflammation ("interstitial"), and alveolitis. F CPQ immunization ameliorated peribronchiolar inflammation. 5plex CPQ immunization did not increase average pathology compared to the mock group. In addition, both F CPQ and 5plex CPQ immunization resulted in lower average lung pathology compared to FI-RSV immunization (Fig. 5H). Histopathological analysis also showed intact alveolar structures, suggesting minimal inflammation in the F CPQ- or 5plex CPQ-immunized group. In contrast, the FI-RSV-immunized group showed marked peribronchial and perivascular inflammation, as the bronchiole walls were markedly thickened, suggesting exaggerated immune response from vaccine-enhanced disease pathology induced by FI-RSV vaccination (Fig. 5I).

DISCUSSION

In this study, we have extended our prior work demonstrating the use of the CoPoP system to display multiple vaccine antigens for a single disease target or viral family, including multiple seasonal influenza strains (14), multiple malaria life-stage targets (20), and multiple alphaviruses (21). Multiple antigens derived from distinct viruses were effectively combined to generate a single vaccine formulation that provided broad protection against various respiratory viruses. Furthermore, the viability of quantifying both total antigen and bound antigen using the ELISA methodology of the 5plexed vaccine was established (fig. S3).

A concurrent surge of seasonal influenza, COVID-19, and RSV cases has emerged as a public health concern that can place strain on healthcare systems (4, 22–24). However, the need for separate vaccinations against each virus contributes to "shot burden," which may hinder vaccine coverage. Accordingly, combination vaccines targeting multiple respiratory viruses could reduce the number of injections while broadening protection. In this study, we developed a three-in-one vaccine formulation and performed a preclinical evaluation to assess its potential to improve protection against these three

major respiratory viruses while reducing the overall vaccination burden during periods of cocirculation.

HA is a primary target for influenza vaccines due to its crucial role in viral entry and established correlates of protection for vaccine-elicited HAI antibodies (25). Serological and in vivo challenge data in mice confirmed that our multivalent approach generated robust neutralizing antibodies and conferred protection against lethal mouse-adapted H1N1, H3N2, and B/Victoria influenza viruses. Weight loss was observed with the highly drifted mouse-adapted A/Hong Kong/1/1968 (H3N2) challenge (Fig. 3). This outcome was likely due to substantial antigenic mismatch, as the vaccine and challenge strain HAs shared only 83% sequence identity (table S1). Frequent antigenic drift of HA is one of the major challenges because it creates a mismatch that allows the virus to evade existing vaccine-induced immunity and, therefore, necessitates the continual updating of seasonal influenza vaccines (26). Interim real-world data from the 2024–2025 season showed that seasonal influenza vaccination provided protection in both outpatient and hospitalization settings, with effectiveness estimates of 32 to 60% and 63 to 78% in children and adolescents, respectively, and 36 to 54% and 41 to 55% in adults (27). Building on the multiplex CoPoP liposomes, this platform is designed for rapid antigen swapping, as recombinant protein-based antigens can potentially be updated and produced faster than traditional egg- or cell-grown vaccine antigens (28). In addition, results in the ferret model demonstrated that the 5plex CPQ elicited influenza-neutralizing antibody levels comparable to those induced by Flublok, a commercial influenza vaccine used here as a benchmark.

When the recombinant RBD was formulated with CPQ for immunization and compared to other contemporary adjuvants, serological assays suggested that 5plex CPQ immunization elicited stronger neutralizing antibodies, most likely due to the benefit of the liposomal antigen presentation (fig. S5). Postvaccination sera efficiently inhibited RBD-ACE2 interaction in the sVNT and neutralized PsV infection in the PsV assays, indicating that the vaccines elicited neutralizing antibodies targeting the RBD, the critical domain that mediates viral entry (29). To further assess protective efficacy, vaccinated K18-hACE2 transgenic mice showed no rSARS-CoV-2-WA1-Nluc signal in the lung. Notably, although no viral titers were detected in the lung homogenates harvested from animals infected with either rSARS-CoV-2-WA1-Nluc or RSV, detectable viral titers were presented in the nasal turbinates. This likely reflects the effect of intramuscular immunization, which predominantly elicits systemic neutralizing antibodies that circulate through the bloodstream and protect internal organs such as the lungs but provide limited immunity in the upper respiratory tract. In contrast, an alternative administration route is intranasal immunization, which primarily induces local mucosal immune responses, including secretory IgA and tissue-resident memory cells within the respiratory mucosa, and has the additional advantage of being a noninvasive and user-friendly route of administration (30).

The FI-RSV vaccine is known to cause vaccine-enhanced disease, also referred to as ERD, which resulted in its clinical failure (19). Poor neutralizing antibody responses were also observed following FI-RSV immunization, likely due to denaturation of neutralizing epitopes (31). In contrast, CPQ correctly presents the neutralizing preF epitope, as confirmed by the recognition of the potent human mAb D25, which specifically recognizes antigenic site Ø exposed on the prefusion F protein (32). No apparent change in average lung pathology was observed in the F CPQ or 5plex CPQ group compared to the mock group (Fig. 5, H and I), indicating a favorable safety

profile. Comparing the two vaccines, histopathological analysis of peribronchiolar, perivascular, and alveolar inflammation showed no apparent worsening of lung pathology in the CPQ group. However, definite exclusion of vaccine-associated ERD requires additional evaluation, such as neutralizing-antibody quality and positive-control vaccine comparison. Due to availability, we only evaluated the 5plex CPQ in vivo protection efficacy against RSV A2 infection in cotton rats. Nevertheless, immunization studies in ICR mice showed that 5plex CPQ elicited neutralizing antibody responses against both RSV A2 and RSV B that were comparable to those induced by the AS01 formulation, underscoring the potential of CPQ for the prevention of RSV infection.

This study prioritized and established the proof of concept that CPQ can serve as a multiplex vaccine platform, as our results demonstrated that the multiplexed formulation conferred immunity against influenza, SARS-CoV-2, and RSV infection in mice, ferrets, and cotton rats challenge models. In this study, AddaVax, AS01, and Alum were used as comparator adjuvant formulations, and CPQ generally induced favorable immunogenicity (fig. S5), although adjuvant dose optimization was not performed. Liposome display, which is only possible with CPQ, nevertheless likely contributed to the improved immunogenicity. Understanding the underlying CPQ immunology mechanism and determining the optimal dose is also critical for future preclinical development of the multiplex CPQ vaccine platform and, therefore, should be examined in the follow-up studies. In addition, all in vivo challenges in the present study involved a single respiratory virus, and whether the protection remains intact against subsequent respiratory pathogen exposure remains for further investigation.

Several limitations should be noted for this study. Although the CoPoP platform elicited protective responses against all five antigens, with no measurable reduction in antigen-specific antibody responses observed within the scope of the assays performed in this study, the prime-boost regimen in naive animals may mask more subtle antigen interference effects under potentially saturating conditions, and additional dose-ranging studies are needed to more definitively assess immune interference. A prior SARS-CoV-2 vaccine based on the CoPoP platform using a His-tagged RBD showed negligible anti-His-tag antibody responses in humans (18). While the use of His-tag could facilitate protein purification, its use is required for the CoPoP platform. The immunogenicity of the trimerization foldon used to generate the HA and F antigens was not reported in these studies. Antigen presentation and immunogenicity of the CPQ-based vaccine largely depend on the structural integrity of the recombinant chimeric proteins used to mimic the viral surface antigen. Also, the use of truncated subunit antigens such as RBD may exclude protective epitopes outside the selected domain, potentially limiting the immune breadth of B cell and T cell responses. Therefore, the efficacy and scalability of a multivalent CPQ-based vaccine depend on efficient antigen selection, design, expression of recombinant proteins, and validation.

In summary, we developed a multivalent vaccine concept on the basis of the CoPoP platform, which can elicit robust protection against multiple respiratory diseases within a single immunization schedule. The flexible formulation of CPQ enables the simple incorporation of multiple protein antigens to target a wide range of respiratory viruses that can be further expanded upon in future studies.

MATERIALS AND METHODS

Materials and vaccine antigens

1,2-Dioleoyl-*sn*-glycero-3-phosphocholine was obtained from Corden-Pharma (catalog no. LP-R4-070). Cholesterol (PhytoChol) was obtained from Wilshire Technologies. CoPoP and PoP were synthesized as previously described (33). Monophosphoryl hexa-acyl lipid A, 3-deacyl (PHAD-3D6A; hereafter PHAD), a synthetic MPLA analog, was obtained from Avanti Research (catalog no. 699855). QS-21 was obtained from Desert King. Trimeric HA ectodomain of A/Victoria/2570/2019 (H1N1) (IQVIA, catalog no. NL2203M), A/Darwin/9/2021 (H3N2) (IQVIA, catalog no. NL2203L), B/Austria/1359417/2021 (B/Victoria lineage) (IQVIA, catalog no. NL2204G), and Protein F from RSV A (strain Long) (in prefusion) (IQVIA, catalog no. NL2308D) were obtained from Nexelis Protein Sciences. Recombinant SARS-CoV-2, S1 subunit protein (RBD) was obtained from RayBiotech Life Inc. (catalog no. 230-30162). Flublok and Fludac for the 2024–2025 influenza season were obtained from McKesson.

Liposome and adjuvant preparation

CoPoP/PHAD liposomes were prepared as previously described and incubated with QS-21 (1 mg ml⁻¹) at a 1:1 mass ratio to PHAD overnight at 4°C (11). The final concentration of CoPoP/PHAD/QS-21 (CPQ) was adjusted to 640 µg ml⁻¹ and stored at 4°C for further use. CPQ (640 µg ml⁻¹) was mixed with each antigen (160 µg ml⁻¹) at a 4:1 mass ratio for 3 hours at room temperature for the 5plex CPQ formulation. Each (0.6 µg) of the five antigens was used to formulate the 5plex CPQ. PoP/PHAD/QS-21 (2HPQ) liposome was prepared in the same way as the CPQ preparation. The AddaVax formulation was prepared by mixing 1:1 of the AddaVax stock solution (InvivoGen, catalog no. vac-adx-10) with PBS containing antigens. AS01 formulation was prepared by adding AS01 liposomes obtained as a component of Shingrix (NDC 58160-823-11) such that the MPLA and QS-21 masses matched the PHAD and QS-21 masses in the CPQ formulation, followed by mixing with antigens. Alum formulation was prepared by diluting the alum stock (VWR, catalog no. 21645-51-2) concentration to 3 mg ml⁻¹ and mixing with an equal amount of antigen with 80 µg of alum per dose. The CPQ liposome and all antigens were sterilized by filtration through a 0.22-µm filter. The final incubation of CPQ liposomes with all antigens was performed in the biosafety cabinet to ensure additional aseptic measures prior to animal administration.

Antigen characterization

Antigen characterization was performed as previously described (34). For SDS-PAGE characterization, antigens were diluted with PBS to 15 µl, and 5× reducing SDS-PAGE sample loading buffer (Thermo Fisher Scientific, catalog no. 39001) was added. The mixture was incubated at 95°C for 5 min. Denatured antigens were loaded into the Nu-PAGE 4 to 12%, Bis-tris, 1.0 to 1.5 mm, Mini Protein Gels (Thermo Fisher Scientific, catalog no. NP0321BOX), and SDS-PAGE was conducted under constant 200 V for 45 min. Gels were washed with deionized (DI) water, gently heated, and shaken for 1 min. The washing and heating step was repeated twice, followed by the addition of Simply Blue SafeStain (Thermo Fisher Scientific, catalog no. LC6065) to fully immerse the gel, gently heated, and stained for 30 min at room temperature with shaking. The gel was destained by washing with DI water, gently heated, and shaken overnight. Antigens were visualized by Bio-Rad ChemiDoc Imager. For antigen cross-linking, 1.5 µg of

each antigen in 10 μ l of DI water was incubated with a serial dilution of bis(succinimidyl) nona(ethylene glycol) (BS(PEG)₉) cross-linker (Thermo Fisher Scientific, catalog no. 21582) to determine the ideal concentration of cross-linker. The cross-linking was continued for 30 min at room temperature and stopped by the addition of 1 M Tris-HCl quenching buffer and further incubated for 15 min. The cross-linked antigens were visualized by SDS-PAGE and the staining method mentioned above. For deglycosylation, 1.5 μ g of each antigen in 10 μ l of DI water was denatured by the addition of dithiothreitol supplied in the Remove-iT PNGase F kit (New England Biolabs, catalog no. P0706) and heated for 5 min at 95°C. The mixture was cooled to room temperature, followed by the addition of Remove-iT PNGase F and further incubated for 1 hour at 37°C. Deglycosylated antigens were visualized by SDS-PAGE and the staining method mentioned above. For anti-His-tag Western blot, 1.5 μ g of each antigen in 15 μ l of PBS was resolved on the gel, followed by the SDS-PAGE method mentioned above, and the antigen was transferred to a 0.2- μ m nitrocellulose membrane (Thermo Fisher Scientific, catalog no. 88013) by the gravity transfer method. The nitrocellulose membrane was blocked by incubation with PBS with 0.05% Tween 20 (PBST), in which PBS containing 5% bovine serum albumin (BSA) (GeminiBio, catalog no. 700-105P), for 2 hours with steady shaking at 37°C. PBST containing 5% BSA and 1:25,000 diluted anti-His-tag antibody (GenScript, catalog no. A00186) for three influenza HA and RBD or 1:1000 diluted Penta-His-tag mAb (Thermo Fisher Scientific, catalog no. P-21315) for F was added to the membrane and further incubated for 1 hour with steady shaking at 37°C. The membrane was washed three times with PBST and incubated with secondary anti-mouse IgG HRP (Cell Signaling Technology, catalog no. 7076) and further incubated for 30 min with steady shaking at 37°C. The membrane was washed three times with PBST, and the chemiluminescence signal was generated with SuperSignal West Dura Extended Duration Substrate (Thermo Fisher Scientific, catalog no. 34076) and captured with Bio-Rad ChemiDoc imager.

Fluorophore labeling of SARS-CoV-2 RBD and quenching assay

Fluorophore labeling of SARS-CoV-2 RBD and subsequent quenching assay were performed as previously described (15). SARS-CoV-2 RBD was first dialyzed in 100 $\times$ 10⁻³ M sodium bicarbonate buffer (pH 9) overnight at 4°C and incubated with DY-490-NHS-Ester (Dyomics, catalog no. DY-490) at a molar ratio of DY-490:RBD at 10:1, with continuous stirring for 1 hour at room temperature in a dark place. The DY-490-labeled RBD was then dialyzed against PBS three times at 4°C for 24 hours to remove the free DY-490. The DY-490-labeled RBD was added to a 96-well plate at a final concentration of 1 μ g/ml alone or with other unlabeled antigens. CPQ or 2HPQ liposomes were added at a mass ratio of 4:1 CoPoP or PoP to total antigens, and the initial fluorescence signal was immediately measured at excitation/emission wavelength of 491/515 nm using the TECAN microplate reader. The plate was kept in a dark place for continuous incubation and fluorescence was measured at different time points. The percentage of DY-490-labeled RBD binding was calculated by

$$\begin{aligned} & \% \text{Antigen binding} \\ &= \left(1 - \frac{\text{Fluorescent intensity}_{\text{liposome+antigen}}}{\text{Fluorescent intensity}_{\text{antigen}}} \right) \times 100\% \end{aligned}$$

Slot blot assay

Total antigen (10 ng) from each group was transferred to a PBS-prewetted nitrocellulose membrane using the 48-well slot blot apparatus (Bio-Rad, catalog no. M1706545). For 5plex CPQ slot blot, the membrane was blocked using 5% BSA in PBS for 1 hour at 37°C and cut into strips, followed by separate incubation with FR494 (100 $\times$), FR1122 (1000 $\times$), and FR1337 (1000 $\times$) from International Reagent Resource, RSV-neutralizing Human (D25) (10,000 $\times$) from ProSci (catalog no. 21-864), and hACE2 (10,000 $\times$) from ACROBiosystems (catalog no. AC2-H52H8) diluted in 5% BSA in PBS for 1 hour at 37°C. The membrane was washed with PBS for 5 min three times, followed by incubation with HRP-conjugated anti-mouse IgG (Cell Signaling Technology, catalog no. 7076) (1000 $\times$) and HRP-conjugated anti-human IgG (Cell Signaling Technology, catalog no. 32935) (10,000 $\times$) for 30 min at 37°C. The membrane was again washed with PBS for 5 min three times and then imaged using the Bio-Rad ChemiDoc Imager.

Ni-NTA competition binding assay

For each 1 μ g of protein, 2 μ l of Ni-NTA magnetic beads (Thermo Fisher Scientific, catalog no. 88831) were washed with PBS, added to the CoPoP liposome sample, and incubated for 30 min at room temperature to ensure the binding of free antigens to the Ni-NTA magnetic beads. The magnetic beads were separated from the supernatant using the magnetic separator (Thermo Fisher Scientific, catalog no. 12321D) after incubation. The beads were resuspended in PBS. A reducing loading sample buffer (5 $\times$) was added to all supernatant and bead resuspension and heated at 95°C for 10 min. All samples were loaded into a 4 to 12% Bis-Tris protein gel for electrophoresis, followed by Coomassie blue staining and imaging with the Bio-Rad ChemiDoc Imager method mentioned above.

Outbred ICR mice immunization

Four- to 6-week-old female ICR mice animal experiments were approved by the University at Buffalo Institutional Animal Care and Use Committee (IACUC; protocol no. BME05044Y). ICR mice received 0.6 μ g of each antigen, which was used to make all formulations as described above. In the 3 $\times$ HA formulation, mice received a total of 1.8 μ g of antigens for each dose, with 7.2 μ g of CoPoP, 2.88 μ g of PHAD, and 2.88 μ g of QS-21. In the F or RBD formulation, mice received a total of 0.6 μ g of antigen for each dose, with 2.4 μ g of CoPoP, 0.96 μ g of PHAD, and 0.96 μ g of QS-21. The 5plex formulation includes all five antigens; mice received a total of 3 μ g of antigens for each dose, with 12 μ g of CoPoP, 4.8 μ g of PHAD, and 4.8 μ g of QS-21. Mice received the intramuscular injections on days 0 and 21. Postimmunized sera were collected on day 47.

Enzyme-linked immunosorbent assay

Recombinant HA ectodomain of A/Brisbane/02/2018 (H1N1) without trimerization domain (eEnzyme catalog no. IA-H1-B18WP), recombinant HA ectodomain of A/Hong Kong/2671/2019 (H3N2) without trimerization domain (eEnzyme, catalog no. IA-H3-HK2671WP), recombinant HA ectodomain of B/Washington/02/2019 (B/Victoria lineage) without trimerization domain (eEnzyme catalog no. IB-HA-W19WP), RSV A2 Fusion glycoprotein ectodomain (SinoBiological, catalog no. 11049-V08B), and recombinant SARS-CoV-2, S1 subunit protein (RBD) (RayBiotech Life, catalog no. 230-30162-1000) were used for ELISA plate coating. ELISA plates were coated with 100 ng of

antigen per well with coating buffer [Na_2CO_3 (5.3 g/liter) and NaHCO_3 (4.2 g/liter) in distilled water, pH 9.6] overnight. ELISA plates were washed three times with PBST, followed by blocking with 2% BSA in PBST with 0.05% Tween 20 solution at 37°C for 1 hour. Vaccinated mice sera or THE His Tag Antibody, mAb, Mouse (catalog no. A00186) was diluted 100× with PBST with 1% BSA, followed by a 10× serial dilution to reach 1.0×10^7 final dilution. The blocking solution was replaced by the serially diluted sera for 1 hour of incubation at 37°C. Plates were washed three times with PBST and incubated with 100 µl of 1000× diluted anti-mouse secondary antibody with HRP (Cell Signaling Technology, catalog no. 7076) per well in PBST with 1% BSA for 30 min at 37°C. Plates were washed with PBST five times, and then 100 µl of BioFX 3,3',5,5'-tetramethylbenzidine (TMB) Conductivity One Component HRP Microwell Substrate (Surmodics, catalog no. 100359-172) was added to each well, followed by the addition of 100 µl of 1 M HCl after 5 min of incubation. The absorbance was measured at 450 nm with the plate reader.

Quantitative ELISA

Total antigen content in CoPoP/PHAD liposome formulations was quantified using a four-parameter logistic (4PL) quantitative ELISA. High-binding 96-well plates were coated with Monoclonal Anti-Influenza A/Wisconsin/67/2022 HA (H1N1) Antibody, Human IgG1 (19G6) (MALS verified) (ACROBiosystems, catalog no. HA1-MY2177) (0.5 µg/ml in carbonate coating buffer, pH 9.6; 100 µl per well) for 1 hour at 37°C, washed three times with PBS containing 0.05% Tween 20 (PBST), and blocked with blocking buffer (2% BSA/PBST; 200 µl per well) for 2 hours at 37°C. For quantification of total antigen, CoPoP liposome formulations were prepared with antigen (40 µg/ml) and treated with 4 M imidazole and 2% Tween 20 at a 1:1:1 volume ratio (sample:imidazole:Tween 20) to release antigen from the particles. Samples were heated at 90°C for 10 min and briefly centrifuged. The released antigen was diluted to 1 µg/ml, followed by a twofold serial dilution series (1.00 to 0.00781 µg/ml) prepared in 1% BSA in PBST. In parallel, antigen standards in PBS (e.g., H1-PBS and 5-plex-PBS) were prepared at the same concentrations to generate calibration curves. Each standard or sample dilution (100 µl) was added to the coated plates and incubated for 1 hour at 37°C, followed by three PBST washes. Then, anti-His-HRP mAb (GenScript, A00612, 1:2000 in diluent; 100 µl per well, 30 min at room temperature) was used for detection. Plates were washed six times with PBST, developed with TMB substrate (100 µl per well, 5 to 10 min at 37°C), and the reaction was stopped with 1 M HCl (100 µl per well). Absorbance at 450 nm was read using a microplate reader. Calibration curves were fitted with a 4PL sigmoidal dose-response model (weighting $1/Y^2$), and sample concentrations (in micrograms per milliliter) were interpolated from the corresponding standard curve.

Free (unbound) antigen in CoPoP formulations was measured using a capture ELISA for His-tagged antigen. High-binding plates were coated with anti-His mAb (GenScript, A00186; 0.5 µg/ml in carbonate coating buffer, pH 9.6; 100 µl per well, 1 hour at room temperature), washed three times with PBST, and blocked with blocking buffer (2% BSA/PBST; 200 µl per well) for 1 hour at room temperature. Antigen standards (H1 or 5-plex in PBS) were diluted to 1.00 µg/ml and diluted in twofold steps to 0.00781 µg/ml (eight-point series, assayed in duplicate). Liposome formulations were prepared with antigen (40 µg/ml) and diluted to 1 µg/ml following by serial dilution to match the standard curve range. Standards or samples (100 µl) were added per well and incubated for 1 hour at room temperature, followed by three PBST washes. For detection of free

H1, plates were incubated Monoclonal Anti-Influenza A (Wisconsin/67/2022) HA (H1N1) Antibody, Human IgG1 (19G6) (MALS verified) (ACROBiosystems, catalog no. HA1-MY2177) with 1:2000 dilution in 1% BSA/PBST (100 µl per well) and incubated 1 hour at room temperature; after incubation, plates were washed three times and then incubated with HRP-conjugated anti-human IgG secondary antibody (1:5000; 100 µl per well, 45 min at room temperature). After six final PBST washes, TMB substrate (100 µl per well) was added for 5 to 10 min, the reaction was stopped with 1 M HCl (100 µl per well), and optical density at 450 nm (OD_{450}) was read within 10 min.

HAI assay

A/California/04/2009 (H1N1) and B/Washington/02/2019 (B/Victoria lineage) influenza viruses were provided by Duke University. A/Switzerland/9715293/2013 (H3N2) influenza virus was provided by Mount Sinai School of Medicine. Vaccinated mice sera (20 µl) were incubated with 80 µl of receptor-destroying enzyme II (RDE II) (Hardy Diagnostics, catalog no. 370013) and 100 µl of PBS at 37°C for 18 hours, followed by incubation at 56°C for 1.5 hours to inactivate RDE II. The sera mixture was twofold serially diluted, and an equal amount of A/California/04/2009 (H1N1), A/Switzerland/9715293/2013 (H3N2), or B/Washington/02/2019 (B/Victoria lineage), diluted with PBS according to the viral titers, was added to the sera mixture and incubated for 30 min at 37°C. Then, 50 µl of 100× diluted red blood cells isolated from turkey whole blood (LAMPIRE Biological Laboratories catalog no. 7209403) was added to the sera/virus mixture. The HAI activity was analyzed after 30 min of incubation at room temperature.

PsV-based neutralization assay

The PsV-based neutralization assay was performed as previously described (15). HEK293/hACE2 cells were cultured in Dulbecco's modified Eagle's medium supplemented with 10% fetal bovine serum (FBS), 1% penicillin-streptomycin, 1 mM sodium pyruvate, and puromycin (2 µg/ml) at 37°C with 5% CO_2 supplement. HEK293/hACE2 cells (1×10^5 per well) were seeded into 96-well plates and incubated overnight. SARS-CoV-2 Pseudoviral Particles, Wuhan-Hu-1 (eEnzyme catalog no. SCV2-PsV-0011), were then incubated with serially diluted sera for 30 min at room temperature and then transferred to the 96-well plate for further 24 hours of incubation with HEK293/hACE2 cells. After incubation, the cells were then washed with PBS and lysed with 30 µl of lysis buffer (Promega, catalog no. E1500) for 10 min at room temperature with gentle shaking. Luciferase substrate (100 µl; CentroPRO catalog no. LB962) was added to each well to measure the luciferase activity.

SARS-CoV-2 sVNT

The SARS-CoV-2 sVNT was performed with the SARS-CoV-2 Surrogate Virus Neutralization Test Kit (GenScript, catalog no. L00847-A) to analyze the RBD-ACE2 interaction inhibition. Briefly, 100× diluted sera and 10× diluted positive and negative controls were incubated with HRP-RBD solution at a 1:1 volume ratio at 37°C for 30 min, and then 100 µl of these samples were transferred to the test plate for an additional 30 min of incubation at 37°C. The plate was washed 5 times, and the percentage of inhibition was calculated by

$$\% \text{Inhibition} = (1 - \text{OD}_{450} \text{ sera} \div \text{OD}_{450} \text{ negative control}) \times 100 \%$$

A cutoff equal to or greater than 30% was used to determine the presence of SARS-CoV-2 neutralizing antibodies according to the manufacturer's protocol.

Mice immunization and in vivo influenza challenge

The BALB/c mice challenge model was approved by the Veterans Administration Western New York Healthcare System's IACUC (protocol no. 509788). Four- to 6-week-old female BALB/c mice received 0.6 µg of each antigen in either 3×HA formulation containing HA of H1N1, H3N2, and B/Victoria lineage or the 5plex formulation containing all five antigens, intramuscularly, followed by a booster on day 21. Mice were intranasally challenged on day 42 after priming with mouse-adapted A/California/04/2009 (H1N1) 1.2×10^5 PFU ($2.5 \times \text{LD}_{50}$), A/Hong Kong/1/1968 (H3N2) 10 PFU ($10 \times \text{LD}_{50}$), or B/Malaysia/2506/2004 (B/Victoria lineage) 3.2×10^6 PFU ($10 \times \text{LD}_{50}$), and the body weight change was monitored for 14 days. Animals were euthanized after 25% weight loss was observed. For clinical measurement, one clinical score point was assigned for each of the following six categories observed: piloerection, labored breathing, hunched posture, lethargy, abnormal gait, and emaciation ($>10\%$ weight loss).

Ferret immunization and H1N1 in vivo challenge

A/Wisconsin/67/2022 (H1N1) in vivo challenge with 10- to 12-week-old Fitch ferrets ($n = 6$ per group, 3 males and 3 females) was carried out by Bioqual Inc. (IACUC protocol no. 23-031). The 5plex formulation included 6 µg of each of the five antigens formulated with 120 µg of CoPoP, 48 µg of PHAD-3D6A, and 48 µg of QS-21. Flublok with an equivalent amount of influenza HA doses in the 5plex formulation was obtained for immunization. Ferrets received 0.5 ml of intramuscular immunization on days 0 and 21, and sera collected were shipped to the University at Buffalo on dry ice for additional serological assays. For Fig. 3G, HAI titers below the limit of detection (LOD) of 10 were assigned “5”; for Fig. 3H, TCID₅₀ titers below the LOD of 2.7 were assigned “1”; for Fig. 3I, microneutralization (MN) titers below the LOD of 8 were assigned “4” for data presentation and statistical analysis.

MN assay for influenza virus

A/Wisconsin/67/2022 (H1N1) influenza virus stocks were propagated in Madin-Darby canine kidney cells. MN assays were performed with minimum essential medium supplemented with trypsin (2 µg/ml), EDTA (1 µg/ml), and gentamicin (50 µg/ml). Ferret sera collected on day 39 postimmunization were serially diluted in twofold steps eight times in assay medium, starting from a 1:8 dilution. An equal volume of virus was added to each dilution to achieve ~400 50% cell culture infectious dose (CCID₅₀)/ml in a final volume of 256 µl. Virus-only controls and cell-only controls were included. Virus-serum mixtures were incubated for 1 hour at room temperature before adding the mixture to 2 wells of 96-well plates with 80 to 100% confluent cells and incubating at 37°C with 5% CO₂ for 4 days for cytopathic effect (CPE) examination. Neutralization titers were reported as the highest reciprocal serum dilution that completely prevented CPE in both replicate wells (complete neutralization). Titers below the lower limit of 8 were assigned a value of “4,” whereas titers above the upper limit of 1024 were assigned “1024” for data presentation.

K18-hACE2 mice immunization

K18-hACE2 mice immunization was performed as previously described (35). All animal studies with a recombinant SARS-CoV-2

WA1 strain expressing Nluc, rSARS-CoV-2–Nluc (WA1), were carried out according to protocols approved by the Texas Biomedical Research Institute at biosafety level 3 IACUC (protocol no. 1718 MU). Female K18-hACE2 transgenic mice (strain no. 034860; Jackson Laboratories, Bar Harbor, ME, USA) ($n = 13$), aged 11 weeks, received intramuscular immunizations on days 0 and 14. Each dose contained 0.6 µg of the following antigens: (i) trimeric HA for influenza H1N1, H3N2, and influenza B (B/Victoria lineage), Prefusion F from RSV A Long (Nexelis Protein Sciences), and recombinant SARS-CoV-2 S1 subunit protein (RBD) (RayBiotech Life Inc.): “5plex”; and (ii) recombinant SARS-CoV-2 S1 subunit protein (RBD) (RayBiotech Life Inc.): “RBD.” These five antigens comprised the 5plex vaccine formulation. Serum samples were collected on 0, 14, and 28 days postvaccination and at 2 and 4 days postinfection and then stored at –20°C to assess serum antibodies against SARS-CoV-2 S protein. Neutralizing titers were determined against infectious SARS-CoV-2 in a plaque reduction neutralization assay. Sera were also sent to the University at Buffalo with dry ice for additional serological assays.

In vivo rSARS-CoV-2–Nluc (WA1) challenge

In vivo rSARS-CoV-2–Nluc (WA1) challenge and infection analysis were performed as previously described (36). To assess protective efficacy, 2 weeks after the second immunization, K18-hACE2 transgenic mice were intranasally inoculated with 10^5 plaque-forming units (PFU) of rSARS-CoV-2–Nluc WA1 (Nluc) isolate (expressing Nluc) in 50 µl of sterile PBS while under anesthesia. Body weight was recorded daily, with the initial weight set as 100%. In survival experiments, mice were monitored every 24 hours. Animals showing lethargy, moribundity, or $>20\%$ loss of initial body weight were humanely euthanized and considered to have succumbed to infection for survival analysis purposes. For the necropsy groups, the nasal turbinate and lungs were collected on days 2 and 4 after the challenge to assess the presence of the virus and to evaluate pathology in the lungs. Mock (PBS)–vaccinated mice challenged with rSARS-CoV-2–Nluc, and unvaccinated and mock-challenged mice were included as internal controls. For in vivo bioluminescence imaging, mice were anesthetized with ketamine and received a 100-µl retro-orbital injection of Nano-Glo luciferase substrate (Promega) and were imaged immediately. Bioluminescence signals were collected and analyzed using Aura software with the AMI Spectrum system. The corresponding scale bar is shown in each figure. Immediately after imaging, the nasal turbinate and lungs were collected and homogenized in 1 ml of PBS. Supernatants were collected, and the presence of SARS-CoV-2 was determined by plaque assay. Nluc activity in the tissue culture supernatants (TCSs) was determined under a multiplate reader (BioTek Instruments Inc.).

SARS-CoV-2 virus titrations

SARS-CoV-2 virus titrations were performed as previously described (35, 36). Nasal turbinate and lungs from SARS-CoV-2–infected mice were homogenized with 1 ml of PBS for 20 s at 7000 rpm using the Precellys tissue homogenizer (Bertin Instruments). Confluent monolayers of Vero AT cells (24-plate format, 4×10^4 cells per well) were infected with 10-fold serial dilutions of TCSs obtained from the nasal turbinate or lung homogenates from SARS-CoV-2–infected mice. The virus from serially diluted samples was adsorbed at 37°C for 1 hour. After viral adsorption, cells were overlaid with postinfection medium containing 1% low-melting agar and were incubated at 37°C for 72 hours. After viral infection, plates were submerged in 10%

neutral buffered formalin (NBF) for 24 hours for fixation/inactivation. Plaque visualization was performed using crystal violet staining. After fixation, the overlay was removed, and cells were washed twice with PBS. Crystal violet solution (0.5% crystal violet in 20% ethanol) was added to the wells and incubated for 30 min. The plates were gently washed with tap water to remove excess stain, and plaques were visualized as clear zones against the stained monolayer.

PRMNT assay for SARS-CoV-2 neutralizing antibodies

PRMNT assays were performed to quantify SARS-CoV-2 neutralizing antibodies as described previously (35). Briefly, serum samples were inactivated at 56°C for 30 min before assay. SARS-CoV-2 (~100 to 200 PFU per well) was mixed with threefold serial dilutions (starting dilution 1:10 for sera samples collected at 14 days post-prime and 1:200 for collected sera at 14 days post-boost) of mouse sera and incubated at 37°C for 1 hour. Confluent monolayers of Vero AT cells (96-well plate format, 4×10^4 cells per well, quadruplicates) were infected with the virus-serum mixture for 1 hour at 37°C. After viral absorption, the infectious medium was exchanged with postinfection medium containing 2% FBS and 1% Avicel. Infected cells were fixed/inactivated 14 hours postinfection with 10% NBF for 24 hours and immunostained with a cross-reactive monoclonal Ab (1C7C7) against the viral nucleocapsid (N) protein of SARS-CoV (1 µg/ml). Viral neutralization was evaluated and quantified using the Enzyme-Linked Immunospot (ELISPOT) assay, and a sigmoidal dose-response, nonlinear regression curve was generated using GraphPad Prism to calculate median neutralization titer in each of the serum samples.

Cotton rat immunization

Cotton rat immunization and RSV A2 viral challenge were performed at Sigmovir Biosystems Inc. (IACUC protocol no. 15). Cotton rats received 1.8 µg of each indicated antigen, with 7.2 µg of CoPoP, 2.88 µg of PHAD, and 2.88 µg of QS-21 per antigen in the formulation. Immunization was performed intramuscularly on days 0 and 21. Sera were collected on days 0, 21, 42, and 47 and shipped to the University at Buffalo on dry ice for additional serological analysis.

RSV IgG ELISA for total RSV or purified F protein

Ninety-six-well plates were coated overnight with ultraviolet-inactivated whole RSV or purified RSV F protein extracted from RSV-infected Hep-2 cells. After decanting the antigen solution, the plates were blocked for 1 hour at room temperature. Plates were washed, and 1:500 diluted sera with duplicates were added to the wells, along with appropriate positive and negative controls, and incubated at room temperature for 1 hour. Following another wash step, a 1:20,000 diluted chicken anti-cotton rat IgG-HRP conjugated (ICL Inc.) was added to each well and incubated for an additional hour at room temperature. After washing, the TMB substrate was added to all wells and incubated for 15 min at room temperature. TMB-stop solution was added to all wells, and the OD₄₅₀ was recorded.

RSV A2 lung and nose viral titration

Lung and nose tissue samples were first homogenized and then clarified by centrifugation. The resulting supernatant was serially diluted in Eagle's Minimum Essential Medium and used to infect confluent Hep-2 cells in 24-well plates in duplicate. After 1 hour of viral adsorption period at 37°C in a 5% CO₂ incubator, the inoculum was removed and replaced with an overlay medium containing 0.75% methylcellulose medium. Following a 4-day incubation, the overlay

was discarded, and the cell monolayers were fixed and stained for 1 hour with 0.1% crystal violet stain. The plates were then rinsed and air-dried, allowing for the counting of plaques, and the viral titer was expressed as plaque-forming units per gram of tissue.

RSV pulmonary histopathology

Lungs were dissected and inflated with 10% NBF to their normal volume, followed by immersion in the same fixative. The lung tissues were embedded in paraffin, sectioned, and stained with hematoxylin and eosin. Slides were then examined blindly to evaluate four parameters of pulmonary inflammation: peribronchiolitis (inflammatory cell infiltration around the bronchioles), perivascularitis (inflammatory cell infiltration around the small blood vessels), interstitial pneumonia (inflammatory cell infiltration and thickening of alveolar walls), and alveolitis (cells within the alveolar spaces). Each parameter was scored for severity on a scale of 0 to 4, and these scores were converted to a 0 to 100% histopathology scale for presentation.

RSV neutralizing antibody assay (60% reduction)

Serum samples were heat inactivated and prepared in a 1:4 ratio serial dilution series after an initial 1:10 dilution in EMEM. The diluted serum was then incubated with 25 to 50 PFU of RSV for 1 hour at room temperature. The virus-serum mixture was inoculated in duplicate onto confluent HEp-2 cell monolayers in 24-well plates. Following a 1-hour adsorption period at 37°C in a 5% CO₂ incubator, the cells were overlaid with 0.75% methylcellulose medium. Plates were then incubated for 4 days, after which the cells were fixed and stained with 0.1% crystal violet stain for 1 hour. After rinsing and air drying, the neutralizing antibody titers are determined as the reciprocal of the serum dilution that achieved a 60% reduction in plaque count relative to the virus-only control.

Real-time PCR detection of RSV

Total RNA was extracted from homogenized lung tissue using the RNeasy purification kit (QIAGEN) and subsequently used to prepare cDNA using SuperScript II RT (Invitrogen) and oligo(dT) primer (Invitrogen). The real-time PCR reaction was performed with Bio-Rad iQ SYBR Green Supermix. The amplifications were performed on a Bio-Rad iCycler for 1 cycle of 95°C for 3 min, followed by 40 cycles of 95°C for 10 s, 60°C for 10 s, and 72°C for 15 s. The iQ5 software (PCR Base Line Subtracted Curve Fit mode) was used to calculate the baseline cycles and cycle threshold (Ct) for each duplicate reaction, and the relative expression quantification was applied to all measurements. The standard curve was created using serial dilutions of a cDNA sample known to be highly enriched for the target transcript (such as the lung tissue from day 4 post-primary RSV infection). The Ct values were plotted against the log₁₀ of the cDNA dilution factor. This curve was then used to convert the Ct values from the experimental samples into relative expression units, and these units were normalized to the expression level of the β-actin housekeeping gene from the corresponding samples.

Cryo-transmission electron microscopy characterization

Samples analyzed by cryo-transmission electron microscopy were vitrified using a Vitrobot Mark IV (Thermo Fisher Scientific). To increase the concentration of liposomes inside the holes in the grids, samples were applied twice to the grids. In the first application, a volume of 3.6 µl was applied to the holey carbon grids and manually blotted using the Vitrobot blotting paper (Standard Vitrobot Filter

Paper, Ø55/20 mm, Grade 595). Right after blotting, a new 3.6-µl drop of the sample was applied to the electron microscopy grid and blotted again using the standard routine with the two blotting pads in the Vitrobot Mark IV for 3 s and with a blot force of +1 before they were plunged into liquid ethane. The Vitrobot was set at 25°C and 100% relative humidity. The grids used for these experiments were C-flat 2/2-2Cu-T, and, before the samples were applied, they were washed with chloroform for 2 hours and treated with negative glow discharge in air at 5 mA for 15 s. Grids were loaded into a Tecnai F20 electron microscope operated at 200 kV using a Gatan 626 single tilt side entry cryo-holder at the Facility for Electron Microscopy Research at McGill. Data acquisition was performed with a TVIPS XF416 CMOS camera using the Serial-EM software (37). All images were collected at a magnification of $\times 62,000$, resulting in images with a calibrated pixel size of $1.761 \text{ \AA}$. Images were collected with a total dose of $\sim 50 \text{ electrons/\AA}^2$ using a defocus ranging from -1.75 to $-2.75 \text{ }\mu\text{m}$.

Statistical analysis

All statistical tests mentioned were performed with GraphPad Prism (version 10.1.1).

Supplementary Materials

This PDF file includes:

Table S1

Figs. S1 to S9

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A subunit vaccine for multiple respiratory viruses

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